Program Schedule and Abstract Book

TEST 2026

May 26 - 27, 2026

This is a synthetic example program. All names, titles, and abstracts are fictitious.

Program Schedule

8:00 AM-9:30 AM, Grand Ballroom, PL1. Plenary Session I, Plenary, Presider: Sophie Jansen, Purdue University

8:00 AM-8:30 AM PL1.1. Demonstration of a cryogenic laser source in LiDAR ranging Liam Wood; Chun Chen

8:30 AM-9:00 AM PL1.2. Thermally-tuned, dissipative beam combiner enabling attosecond science Oliver Ryan

9:00 AM-9:30 AM PL1.3. Demonstration of a monolithic optomechanical oscillator in coherent detection Camilla Marino

9:45 AM-11:45 AM, W101, TM1A. Integrated Photonic Devices, A&T Oral, Applications & Technology 3, Presider: Robert WALKER

9:45 AM-10:15 AM TM1A.1. [Invited Talk] Demonstration of a wavelength-agile waveguide lattice in photonic computing Ma Ling; Tang Kun; Gao Feng; Zhou Na; Hui Cheng

10:15 AM-10:45 AM TM1A.2. Scalable, spatially-multiplexed interferometer enabling beam steering Peng Peng; Peng Bo; He Yan; Liu Yan; Ming Tseng; Ting Ho; He Jun

10:45 AM-11:15 AM TM1A.3. Topological supercontinuum for quantum key distribution Maryam Zaki; Gabriele Rossi; Davide Fontana; Simone Rizzo; Callum Robinson

11:15 AM-11:45 AM TM1A.4. Toward broadband mid-infrared generation using a gain medium Maeda Takuya; Yamamoto Tomoya; Aoife MacLeod; Johanna Wagner; Goto Daiki

9:45 AM-11:45 AM, W102, TM1B. Quantum Light Sources, S&I Oral, Science and Innovations 5, Presider: Charlotte Jones, Los Alamos

9:45 AM-10:15 AM TM1B.1. [Invited Talk] Nonlinear quantum emitter for terahertz imaging Zhu Feng; Huang Hua; Zhang Jun; Ma Kun; Chih Chiang; Valentina Russo; Lin Rui

10:15 AM-10:45 AM TM1B.2. Dispersion-engineered, quantum-limited microresonator enabling attosecond science Shimizu Naoki; Mori Naoki; Tanaka Daiki; Maeda Hiroshi; Fatima Saleh; Yoshida Yui

10:45 AM-11:15 AM TM1B.3. Broadband, polarization-resolved thin-film platform enabling beam steering Sara Al-Saud

11:15 AM-11:45 AM TM1B.4. Abstract Withdrawn

9:45 AM-11:45 AM, W103, TM1C. Ultrafast Phenomena, FS Oral, Fundamental Science 2, Presider: Yamamoto Kenji, Tohoku University

9:45 AM-10:15 AM TM1C.1. [Invited Talk] Demonstration of a wavelength-agile laser source in quantum key distribution Liu Kun; Omar Mahmoud; Jang Joon; Wei Wu; Wen Chou; Hao Lai

10:15 AM-10:45 AM TM1C.2. Mode-locked, monolithic modulator enabling biophotonics Hao Yang; Liu Tao; Ming Ho; Li Qiang; Ming Yeh

10:45 AM-11:15 AM TM1C.3. Self-referenced spectrometer for precision timing Cho Jiho; Choi Seoyeon; Kim Yuna; Seo Jiwoo; Yoo Yuna; Seo Jiwoo; Hwang Joon; Choi Woojin

11:15 AM-11:45 AM TM1C.4. Dissipative interferometer for terahertz imaging Dimitrios Athanasiou; Dimitrios Papadopoulos; Aoife Wood; Eleni Oikonomou; Noor de Jong; Georgia Makris; Liam O'Brien; Christos Georgiou

9:45 AM-11:45 AM, W104, TM1D. Nonlinear Frequency Conversion, S&I Oral, Science and Innovations 8, Presider: Olga Golubev, University of Regensburg

9:45 AM-10:15 AM TM1D.1. [Invited Talk] Toward resonant LiDAR ranging using a ring resonator Ishikawa Ryo

10:15 AM-10:45 AM TM1D.2. Demonstration of a high-finesse waveguide lattice in coherent detection Aisha Younis; Ali Al-Saud; Ciara Davies; Sanne Dekker; Ewan Hall; Callum Patterson

10:45 AM-11:15 AM TM1D.3. Toward polarization-resolved optical networking using a photodetector Arjun Shah; Peng Peng; Zhang Wei

11:15 AM-11:45 AM TM1D.4. Demonstration of a ultrafast optical cavity in photonic computing Joshua Jackson; Adrian Rodriguez; Niall Smith; Michal Kiss; Amelia Thomas; Jack Wood; Jonas Meyer; Andrea Ramos

1:30 PM-3:30 PM, W101, TM2A. Optical Sensing Systems, A&T Oral, Applications & Technology 3, Presider: YAMAMOTO Sakura

1:30 PM-2:00 PM TM2A.1. [Invited Talk] Ultrafast, dispersion-engineered Bragg grating enabling frequency metrology Zaid Al-Farsi; Lin Qiang; Andrea Fernandez; Isabel Moreno; Thomas Green; Charlotte Fraser; Omar Saleh; Xie Shan

2:00 PM-2:30 PM TM2A.2. Demonstration of a reconfigurable feedback loop in optical networking Xu Zhen; Lin Feng; Lachlan Green; Deng Fang; Hao Yang; Amira Habib; Chen Yang; Feng Feng

2:30 PM-3:00 PM TM2A.3. Demonstration of a electrically-pumped modulator in neuromorphic computing Amelia Wright; Liam Wright; Grace Wilson; Vladimir Smirnov; Hu Bo

3:00 PM-3:30 PM TM2A.4. Demonstration of a high-finesse interferometer in attosecond science Gabriele Ricci

1:30 PM-3:30 PM, W102, TM2B. Laser Resonator Design, S&I Oral, Science and Innovations 5, Presider: Oliver Murphy, Air Force Research Laboratory

1:30 PM-2:00 PM TM2B.1. [Invited Talk] Demonstration of a tunable laser source in remote sensing Okada Emi; Kato Naoki; Kimura Sho

2:00 PM-2:30 PM TM2B.2. Cryogenic interferometer for frequency metrology William Hall

2:30 PM-3:00 PM TM2B.3. Toward spatially-multiplexed LiDAR ranging using a fiber amplifier Florian Lehmann; Tamar Biton

3:00 PM-3:30 PM TM2B.4. Programmable, low-noise nanobeam enabling neuromorphic computing Yoon Soyeon; Khalid Farouk; Kim Jaewon; Park Jisoo; Jang Hana; Ahn Jihun; Lim Soyeon; Justin Garcia

1:30 PM-3:30 PM, W103, TM2C. Coherent Optical Networks, FS Oral, Fundamental Science 2, Presider: Gregory Davis, Purdue University

1:30 PM-2:00 PM TM2C.1. [Invited Talk] Chip-scale, programmable optomechanical oscillator enabling LiDAR ranging Zhao Na

2:00 PM-2:30 PM TM2C.2. Toward gain-switched quantum sensing using a photonic crystal Dimitra Christodoulou; Sara Ricci; Rachel Carter

2:30 PM-3:00 PM TM2C.3. Low-noise, tunable laser source enabling optical networking Henry Wright; Hao Kuo; Mary Carter; Freya Wood; Sara Gallo; Saoirse Kelly; Isabel Diaz; Ruby Robinson

3:00 PM-3:30 PM TM2C.4. Toward few-cycle remote sensing using a metasurface Ramesh Agarwal; Hassan Al-Farsi; Sarah Martin; Aarav Desai

1:30 PM-3:30 PM, W104, TM2D. Mid-Infrared Technologies, S&I Oral, Science and Innovations 8, Presider: Elena Vasilyev, University of Central Florida

1:30 PM-2:00 PM TM2D.1. [Invited Talk] Demonstration of a dissipative photodetector in frequency metrology Yoo Seojun; Kim Hyeon

2:00 PM-2:30 PM TM2D.2. Toward self-referenced frequency metrology using a photodetector Matthew Thompson; Mario Gil; William Ryan; Lily Hughes; Gilad Friedman; Ava Williams; Ava McCarthy; Megan Wilson

2:30 PM-3:00 PM TM2D.3. Wavelength-agile, programmable interferometer enabling frequency metrology Federico Lombardi

3:00 PM-3:30 PM TM2D.4. Toward scalable precision timing using a interferometer Ma Jing; Hu Min; Chun Cheng; Huang Zhen; Gao Yu; Sun Hao; Sun Qiang

4:00 PM-5:30 PM, W105, SYMP1. Symposium on Emerging Photonic Materials, Symposium, Special Symposia, Presider: Charles Anderson

4:00 PM-4:30 PM SYMP1.1. [Invited Talk] Toward ultrafast attosecond science using a thin-film platform Kai Hsu

4:30 PM-5:00 PM SYMP1.2. [Invited Talk] Broadband gain medium for biophotonics Goto Satoshi; Watanabe Satoshi; Zhao Hao; Watanabe Yui; Hashimoto Yusuke; Kimura Yui; Yousef Mansour

5:00 PM-5:30 PM SYMP1.3. [Invited Talk] Scalable quantum emitter for spectroscopy Wen Tseng; Shin Donghyun; Patricia Carter; Shin Jihun; Suresh Chatterjee

5:30 PM-7:00 PM, Exhibit Hall, JTUP. Poster Session I, Poster

5:30 PM-7:00 PM JTUP.1. Hybrid, dissipative quantum emitter enabling optical communication Zhou Feng

5:30 PM-7:00 PM JTUP.2. Toward hybrid photonic computing using a feedback loop Declan Kelly

5:30 PM-7:00 PM JTUP.3. Wavelength-agile, gain-switched entangled-photon pair enabling optical communication Suzuki Satoshi

5:30 PM-7:00 PM JTUP.4. Spatially-multiplexed, single-photon microresonator enabling remote sensing Freddie McCarthy; Liam Jones; Sanne Wouters; Ting Liu

5:30 PM-7:00 PM JTUP.5. Low-noise Bragg grating for mid-infrared generation Ananya Mukherjee

5:30 PM-7:00 PM JTUP.6. Toward low-noise quantum key distribution using a nanobeam Riccardo Galli; Luca Rinaldi

5:30 PM-7:00 PM JTUP.7. Dispersion-engineered thin-film platform for mid-infrared generation Antonio Leone; Bram Claes; Faisal Rahman; William Smith; Luca Ferrari

5:30 PM-7:00 PM JTUP.8. Toward self-referenced neuromorphic computing using a gain medium Wu Bin; Peng Kun

5:30 PM-7:00 PM JTUP.9. Quantum-limited, scalable fiber amplifier enabling optical networking Oliver Robinson; Oliver Patterson

5:30 PM-7:00 PM JTUP.10. Coherent, spatially-multiplexed saturable absorber enabling beam steering Freya Taylor; Elias Werner

5:30 PM-7:00 PM JTUP.11. Toward chip-scale attosecond science using a supercontinuum Ahmed Farouk; Bram Claes

5:30 PM-7:00 PM JTUP.12. Gain-switched, high-finesse waveguide lattice enabling mid-infrared generation Varun Patel; Neha Naidu; Sanjay Gupta

8:30 AM-10:30 AM, W101, WM1A. Photonic Computing Architectures, A&T Oral, Applications & Technology 3, Presider: LI Jun, CU Boulder

8:30 AM-9:00 AM WM1A.1. [Invited Talk] Demonstration of a electrically-pumped metasurface in coherent detection Yousef Habib; Freddie Thompson; Agnieszka Novak; Anita Kaminski; Saoirse Evans; Istvan Szymanski; Federico Greco

9:00 AM-9:30 AM WM1A.2. Demonstration of a coherent photonic crystal in biophotonics Faisal Habib; Robert Martinez; Lucia Ruiz; Ma Lei; Liang Ping; Hu Feng; Sanne van Dijk; Faisal Saleh

9:30 AM-10:00 AM WM1A.3. Toward wavelength-agile biophotonics using a interferometer Grace Jones; Ruby Wood; Thomas Patterson; Rocio Diaz; Pieter Vermeulen; Kevin Phillips; Maryam Hassan

10:00 AM-10:30 AM WM1A.4. Demonstration of a thermally-tuned nanobeam in mid-infrared generation Tang Yu; Han Hui; Cheng Lai; Okada Yui; Nakamura Haruki; Shu Chang

8:30 AM-10:30 AM, W102, WM1B. Microresonator Combs, S&I Oral, Science and Innovations 5, Presider: Abe Tomoya

8:30 AM-9:00 AM WM1B.1. [Invited Talk] Thermally-tuned optical cavity for quantum key distribution Hong Chang; Riccardo Mancini; Pei Kuo; Ma Fang; Xie Peng; Jia Lee; Miguel Reyes; Wen Yang

9:00 AM-9:30 AM WM1B.2. Resonant, monolithic ring resonator enabling attosecond science Lily Thompson; Allison Jones; Charlotte O'Brien; Bilal Said; Linda Carter

9:30 AM-10:00 AM WM1B.3. Self-referenced interferometer for coherent detection Yousef Said; Suzuki Sota; Thomas Davies; Huang Tao; Eitan Adler; Feng Yan; Lukas Kovacs; Mary Anderson

10:00 AM-10:30 AM WM1B.4. Abstract Withdrawn

8:30 AM-10:30 AM, W103, WM1C. Fiber Laser Advances, FS Oral, Fundamental Science 2, Presider: Ming Chen, Univ of Arizona, Coll of Opt Sciences

8:30 AM-9:00 AM WM1C.1. [Invited Talk] Scalable, reconfigurable quantum emitter enabling coherent detection Allison Baker

9:00 AM-9:30 AM WM1C.2. Demonstration of a quantum-limited ring resonator in optical communication Declan Green; Niall Murphy; Ma Long; Yu Wu; Daniel Phillips; Nikolaos Spanos; Eric Clark; Bilal Aziz

9:30 AM-10:00 AM WM1C.3. Toward ultrafast neuromorphic computing using a phased array Sigrid Lund

10:00 AM-10:30 AM WM1C.4. Topological nanobeam for mid-infrared generation Fujita Hiroshi; Nakamura Rina

8:30 AM-10:30 AM, W104, WM1D. Metasurface Engineering, S&I Oral, Science and Innovations 8, Presider: Oscar Hughes, University of Sydney

8:30 AM-9:00 AM WM1D.1. [Invited Talk] Dispersion-engineered, phase-stable optical cavity enabling biophotonics Omar Younis; Divya Singh

9:00 AM-9:30 AM WM1D.2. Coherent, resonant spectrometer enabling mid-infrared generation Agnieszka Molnar; Isla Brown; Ville Sandberg; Elias Frank; William Thompson; Wouter Aerts; Daan van Dijk

9:30 AM-10:00 AM WM1D.3. Demonstration of a programmable gain medium in spectroscopy Peng Feng; Deng Yan

10:00 AM-10:30 AM WM1D.4. Demonstration of a resonant ring resonator in biophotonics Francesco Galli

10:30 AM-12:00 PM, W101, WM2A. Optical Systems, A&T Oral, Applications & Technology 3, Presider: ISHIKAWA Akira, Tamagawa University

10:30 AM-11:00 AM WM2A.1. Electrically-pumped beam combiner for coherent detection Sasaki Ryo; Yamaguchi Aoi; Jia Yang; Okada Yuki; Kimura Keisuke

11:00 AM-11:30 AM WM2A.2. Demonstration of a scalable waveguide lattice in LiDAR ranging Michal Dvorak; Feng Zhen; Carlos Fernandez; Giuseppe Lombardi; Lorenzo Romano; Ewan Williams; Florian Dubois; Dimitra Vlachos

11:30 AM-12:00 PM WM2A.3. Demonstration of a nonlinear interferometer in precision timing Pei Lee; Hao Yang; Wang Yang; Zhu Bin; Hsin Wu; Wen Yang; Deng Lei

10:30 AM-12:00 PM, W102, WM2B. Optical Systems, S&I Oral, Science and Innovations 5, Presider: Klara Bauer, Univ degli Studi di Roma La Sapienza

10:30 AM-11:00 AM WM2B.1. Spatially-multiplexed, thermally-tuned optomechanical oscillator enabling beam steering Arjun Menon; Rohan Kumar; Aarav Rao

11:00 AM-11:30 AM WM2B.2. Reconfigurable entangled-photon pair for attosecond science Sem de Boer; Jack Thomas; Sophie Green; Mary Thomas; Liam Robinson; Sophie Thomas; Davide Barbieri; Niall Williams

11:30 AM-12:00 PM WM2B.3. Demonstration of a nonlinear optical cavity in spectroscopy Xie Ping; Feng Bo; He Lei; Jia Chiang; Feng Yang; Ting Liu; Zhang Long; Sun Fang

10:30 AM-12:00 PM, W103, WM2C. Optical Systems, FS Oral, Fundamental Science 2, Presider: Chih Tsai, University of Southern California

10:30 AM-11:00 AM WM2C.1. Demonstration of a thermally-tuned photonic crystal in quantum sensing Kwon Jiwoo

11:00 AM-11:30 AM WM2C.2. Self-referenced, resonant beam combiner enabling frequency metrology Wang Jing; Gao Hui; Alice Moretti; Liam Wright; Yu Lai; Ahmed Khalil; Yousef Nasser

11:30 AM-12:00 PM WM2C.3. Toward topological nonlinear microscopy using a phased array Amelia Edwards; Li Ping; Zhou Long; Olga Kozlov; Joshua Anderson; Richard Taylor; Huang Shan; Jiri Marek

10:30 AM-12:00 PM, W104, WM2D. Optical Systems, S&I Oral, Science and Innovations 8, Presider: Andrew Walker

10:30 AM-11:00 AM WM2D.1. Demonstration of a broadband nanobeam in quantum key distribution Patricia Young; Thomas Wood; Rocio Rodriguez; Amira Younis; Meera Choudhury; Ivan Pavlov

11:00 AM-11:30 AM WM2D.2. Demonstration of a resonant feedback loop in attosecond science Tariq Ibrahim; Cai Kun; Cai Long; Chih Wu; Wang Jing; Han Jing

11:30 AM-12:00 PM WM2D.3. Toward high-finesse biophotonics using a frequency comb Elena Alonso; Sophie White; Lieke Peeters

1:30 PM-3:00 PM, W101, WT1. Tutorials in Quantum Photonics, S&I Oral, Science and Innovations 9, Presider: Michael Lewis, Max-Planck-Inst Physik des Lichts

1:30 PM-2:00 PM WT1.1. [Tutorial Talk] Nonlinear supercontinuum for photonic computing Nathan Wright; Aoife Robinson; Lukas Bauer

2:00 PM-2:30 PM WT1.2. Demonstration of a mode-locked spectrometer in photonic computing Freddie MacLeod

2:30 PM-3:00 PM WT1.3. Toward chip-scale optical communication using a optomechanical oscillator Noura Rahman; Lily Evans; Matthew Jones; Olivia Brown; Kabir Bhat

3:30 PM-4:30 PM, W101, PD1. Postdeadline Session 1, Postdeadline Oral, Presider: Kai Cheng, Oak Ridge National Laboratory

3:30 PM-4:00 PM PD1.1. Toward coherent photonic computing using a Bragg grating Guo Shan; Pei Liu; Luo Tao; Yu Lai; Xu Hui; Gao Yan; Zhang Wei

4:00 PM-4:30 PM PD1.2. Few-cycle, broadband spectrometer enabling LiDAR ranging Callum Davies; Jonas Braun; Grace Thompson

3:30 PM-4:30 PM, W102, PD2. Postdeadline Session 2, Postdeadline Oral, Presider: Hong Wu, University of Maryland

3:30 PM-4:00 PM PD2.1. Gain-switched, nonlinear ring resonator enabling beam steering Noura Nasser; Zhu Yan; Liang Kun; Simone Conti; Fatima Saleh

4:00 PM-4:30 PM PD2.2. Low-noise phased array for frequency metrology Megan Rodriguez; Lorenzo Mancini; Stephen Baker

Final ID: PL1.1

Demonstration of a cryogenic laser source in LiDAR ranging

Liam Wood;[1]; *Chun Chen;*[1];
1. Oak Ridge National Laboratory, Oak Ridge, TN, United States.

Abstract (35 Word Limit): We present measurements of a low-noise waveguide lattice operating in the context of beam steering. The device achieves competitive performance across the operating band. These findings motivate further study of the underlying dynamics.

Final ID: PL1.2

Thermally-tuned, dissipative beam combiner enabling attosecond science

Oliver Ryan;[1,2];
1. Applied Physics, Stanford University, Mountain View, CA, United States.
2. SLAC National Accelerator Laboratory, Menlo Park, CA, United States.

Abstract (35 Word Limit): We report a cryogenic ring resonator that accelerates remote sensing, reaching a value below 5³ cm/decade in our setup. The measured figures of merit exceed prior comparable demonstrations. The measured figures of merit exceed prior comparable demonstrations.

Final ID: PL1.3

Demonstration of a monolithic optomechanical oscillator in coherent detection

Camilla Marino;[1];
1. Department of Electrical Engineering and Computer Science, Massachusetts Institute of Technology, Boston, MA, United States.

Abstract (35 Word Limit): This work introduces a reconfigurable supercontinuum as a platform for LiDAR ranging. Results indicate a clear pathway toward practical deployment.

Final ID: TM1A.1

Demonstration of a wavelength-agile waveguide lattice in photonic computing

Ma Ling;[1]; *Tang Kun;*[1]; *Gao Feng;*[1]; *Zhou Na;*[1]; *Hui Cheng;*[1];
1. The Hong Kong University of Science and Technology (Guangzhou), Guangzhou, Guangdong, China.

Abstract (35 Word Limit): This work introduces a quantum-limited beam combiner as a platform for remote sensing. These findings motivate further study of the underlying dynamics.

Final ID: TM1A.2

Scalable, spatially-multiplexed interferometer enabling beam steering

Peng Peng;[1]; *Peng Bo;*[1,2]; *He Yan;*[1]; *Liu Yan;*[1,2]; *Ming Tseng;*[1]; *Ting Ho;*[1]; *He Jun;*[1];
1. School of Microelectronics, South China University of Technology, Guangzhou, China.
2. School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore, Singapore.

Abstract (35 Word Limit): A wavelength-agile spectrometer is demonstrated for terahertz imaging. We discuss the trade-offs governing the design and their implications. We discuss the trade-offs governing the design and their implications.

Final ID: TM1A.3

Topological supercontinuum for quantum key distribution

Maryam Zaki;[1,2]; *Gabriele Rossi;*[3]; *Davide Fontana;*[1,2]; *Simone Rizzo;*[1,2]; *Callum Robinson;*[1,2];
1. Hybrid photonic laboratory, EPFL, 1015-CH, Switzerland, Lausanne, Switzerland.
2. Center for Quantum Science and Engineering, EPFL, 1015-CH, Switzerland, Lausanne, Switzerland.
3. Department of Chemistry and Industrial Chemistry, University of Genoa, 16146 Genova, Italy, Genova, Italy.

Abstract (35 Word Limit): We present measurements of a few-cycle frequency comb operating in the context of optical networking. The device achieves competitive performance across the operating band.

Final ID: TM1A.4

Toward broadband mid-infrared generation using a gain medium

Maeda Takuya;[1]; *Yamamoto Tomoya;*[1]; *Aoife MacLeod;*[1,2]; *Johanna Wagner;*[3]; *Goto Daiki;*[1];

1. Department of Physics, Institute of Science Tokyo, Meguro-ku, Tokyo, Japan.
2. Institute of industrial Science, The Univerce of Tokyo, 4-6-1 Komaba, Meguro-ku, Tokyo, Japan.
3. School of Engineering and Technology, Central Michigan University, Mount Pleasant, MI, United States.

Abstract (35 Word Limit): We report a coherent entangled-photon pair that accelerates photonic computing. We discuss the trade-offs governing the design and their implications.

Final ID: TM1B.1

Nonlinear quantum emitter for terahertz imaging

Zhu Feng;[1]; *Huang Hua;*[1]; *Zhang Jun;*[2]; *Ma Kun;*[2]; *Chih Chiang;*[1]; *Valentina Russo;*[1]; *Lin Rui;*[1];
1. Univ of Electronic Science & Tech China, Chengdu, SICHUAN, China.
2. Xidian University, Xi'an, China.

Abstract (35 Word Limit): We report a monolithic interferometer that accelerates precision timing. The measured figures of merit exceed prior comparable demonstrations.

Final ID: TM1B.2

Dispersion-engineered, quantum-limited microresonator enabling attosecond science

Shimizu Naoki;[1]; *Mori Naoki;*[1]; *Tanaka Daiki;*[1]; *Maeda Hiroshi;*[1]; *Fatima Saleh;*[1]; *Yoshida Yui;*[1];
1. Graduate School of Engineering, Mie University, Tsu, Mie, Japan.

Abstract (35 Word Limit): A broadband quantum emitter is demonstrated for quantum sensing, reaching a value below 5⁻⁹ Hz/decade in our setup. The device achieves competitive performance across the operating band. Our approach reduces system complexity while preserving fidelity.

Final ID: TM1B.3

Broadband, polarization-resolved thin-film platform enabling beam steering

Sara Al-Saud;[1];
1. University of Arizona, Tucson, AZ, United States.

Abstract (35 Word Limit): A single-photon entangled-photon pair is demonstrated for frequency metrology, reaching a value below 2³ cm/decade in our setup. These findings motivate further study of the underlying dynamics. The measured figures of merit exceed prior comparable demonstrations.

Final ID: TM1C.1

Demonstration of a wavelength-agile laser source in quantum key distribution

Liu Kun;[1]; *Omar Mahmoud;*[1]; *Jang Joon;*[1]; *Wei Wu;*[2]; *Wen Chou;*[3]; *Hao Lai;*[1];
1. UCLA, Los Angeles, CA, United States.
2. State Key Laboratory of Functional Materials for Informatics, Shanghai Institute of Microsystem and Information Technology, Shanghai, China.
3. Institute of Microelectronics, A*STAR, Singapore, Singapore.

Abstract (35 Word Limit): A topological saturable absorber is demonstrated for remote sensing. The device achieves competitive performance across the operating band.

Final ID: TM1C.2

Mode-locked, monolithic modulator enabling biophotonics

Hao Yang;[1]; *Liu Tao;*[1]; *Ming Ho;*[1]; *Li Qiang;*[1]; *Ming Yeh;*[1];
1. The Hong Kong University of Science and Technology (Guangzhou), Guangzhou, Guangdong, China.

Abstract (35 Word Limit): This work introduces a dispersion-engineered supercontinuum as a platform for remote sensing. These findings motivate further study of the underlying dynamics.

Final ID: TM1C.3

Self-referenced spectrometer for precision timing

Cho Jiho;[1,2]; *Choi Seoyeon;*[1]; *<u>Kim Yuna;</u>*[1]; *Seo Jiwoo;*[1,2]; *Yoo Yuna;*[1,2]; *Seo Jiwoo;*[1,3]; *Hwang Joon;*[1,5]; *Choi Woojin;*[1,4];

1. Center for Quantum Technology, Korea Institute of Science and Technology (KIST), Seoul, Korea (the Republic of).
2. School of Electrical Engineering, Korea University, Seoul, Korea (the Republic of).
3. Department of Electrical and Computer Engineering, Ajou University, Suwon, Korea (the Republic of).
4. Department of Physics, Kyung Hee University, Seoul, Korea (the Republic of).
5. Division of Quantum Information, KIST School, Korea University of Science and Technology (KIST), Seoul, Korea (the Republic of).

Abstract (35 Word Limit): Here we investigate a thermally-tuned waveguide lattice suitable for remote sensing. We discuss the trade-offs governing the design and their implications.

Final ID: TM1C.4

Dissipative interferometer for terahertz imaging

<u>Dimitrios Athanasiou;</u>[1]; *Dimitrios Papadopoulos;*[1]; *Aoife Wood;*[1]; *Eleni Oikonomou;*[1]; *Noor de Jong;*[2]; *Georgia Makris;*[1]; *Liam O'Brien;*[2]; *Christos Georgiou;*[1];
1. INSTITUTE OF COMMUNICATION & COMPUTER SYSTEMS (ICCS) , National Technical University of Athens (NTUA), Athens, Greece.
2. Ghent University and IMEC, 2Center for Microsystem Technology (CMST), Ghent, Belgium.

Abstract (35 Word Limit): We report a chip-scale metasurface that supports photonic computing, reaching a value below 5³ s/decade in our setup. We discuss the trade-offs governing the design and their implications. The device achieves competitive performance across the operating band.

Final ID: TM1D.1

Toward resonant LiDAR ranging using a ring resonator

Ishikawa Ryo;[1];
1. University of Electro-Communications, Chofu, Japan.

Abstract (35 Word Limit): This work introduces a polarization-resolved photodetector as a platform for terahertz imaging. These findings motivate further study of the underlying dynamics. Results indicate a clear pathway toward practical deployment.

Final ID: TM1D.2

Demonstration of a high-finesse waveguide lattice in coherent detection

Aisha Younis;[2]; *Ali Al-Saud;*[1]; *Ciara Davies;*[1,2]; *Sanne Dekker;*[1]; *Ewan Hall;*[1,2]; *Callum Patterson;*[1];
1. CREOL, College of Optics and Photonics, University of Central Florida, Orlando, FL, United States.
2. City University of New York, New York, NY, United States.

Abstract (35 Word Limit): A single-photon quantum emitter is demonstrated for spectroscopy. The device achieves competitive performance across the operating band. Characterization confirms agreement with the proposed model.

Final ID: TM1D.3

Toward polarization-resolved optical networking using a photodetector

Arjun Shah;[1]*;* *Peng Peng;*[1]*;* *Zhang Wei;*[1]*;*
1. Baylor University, Waco, TX, United States.

Abstract (35 Word Limit): A mode-locked nanobeam is demonstrated for remote sensing. Results indicate a clear pathway toward practical deployment. Our approach reduces system complexity while preserving fidelity.

Final ID: TM1D.4

Demonstration of a ultrafast optical cavity in photonic computing

Joshua Jackson;[1]; *Adrian Rodriguez;*[1]; *Niall Smith;*[2]; *Michal Kiss;*[2]; *Amelia Thomas;*[1]; *Jack Wood;*[2]; *Jonas Meyer;*[2]; *Andrea Ramos;*[1];

1. UCLA, Los Angeles, CA, United States.

2. DESY, Hamburg, Germany.

Abstract (35 Word Limit): A spatially-multiplexed photodetector is demonstrated for photonic computing. Characterization confirms agreement with the proposed model.

Final ID: TM2A.1

Ultrafast, dispersion-engineered Bragg grating enabling frequency metrology

Zaid Al-Farsi;[1]; *Lin Qiang;*[1]; *Andrea Fernandez;*[1]; *Isabel Moreno;*[2]; *Thomas Green;*[1,3]; *Charlotte Fraser;*[1,4]; *Omar Saleh;*[5]; *Xie Shan;*[5,6];

1. Harvard University, Cambridge, MA, United States.
2. Department of Physics, The University of Texas at Austin, Austin, TX, United States.
3. Institute of Experimental Physics, Graz University of Technology, Graz, Styria, Austria.
4. Institute of Nanotechnology, Karlsruhe Institute of Technology, Karlsruhe, Baden-Wurttemberg, Germany.
5. Department of Chemical Engineering and Materials Science, University of California, Irvine, Irvine, CA, United States.
6. Department of Physics and Astronomy, and Irvine Materials Research Institute (IMRI), University of California, Irvine, Irvine, CA, United States.

Abstract (35 Word Limit): This work introduces a scalable optomechanical oscillator as a platform for biophotonics. The device achieves competitive performance across the operating band.

Final ID: TM2A.2

Demonstration of a reconfigurable feedback loop in optical networking

Xu Zhen;[1]; *Lin Feng;*[2]; *Lachlan Green;*[3]; *Deng Fang;*[1]; *Hao Yang;*[4]; *Amira Habib;*[3]; *Chen Yang;*[2]; *Feng Feng;*[5];

1. Department of Electrical and Computer Engineering, University of Wisconsin-Madison, Madison, WI, United States.
2. Department of Electrical and Computer Engineering, National University of Singapore, Singapore, Singapore, Singapore.
3. Department of Electrical and Microelectronic Engineering, Rochester Institute of Technology, Rochester, NY, United States.
4. Department of Electrical Engineering, University of Arkansas, Fayetteville, AR, United States.
5. The University of Texas at Dallas, Richardson, TX, United States.

Abstract (35 Word Limit): This work introduces a self-referenced photodetector as a platform for nonlinear microscopy. The measured figures of merit exceed prior comparable demonstrations.

Final ID: TM2A.3

Demonstration of a electrically-pumped modulator in neuromorphic computing

<u>Amelia Wright;</u>[1]; *Liam Wright;*[2]; *Grace Wilson;*[1]; *Vladimir Smirnov;*[3]; *Hu Bo;*[1];
1. Stanford University, Stanford, CA, United States.
2. Harvard University, Cambridge, MA, United States.
3. Weizmann Institute of Science, Rehovot, Israel.

Abstract (35 Word Limit): We report a low-noise feedback loop that stabilizes quantum key distribution. Our approach reduces system complexity while preserving fidelity.

Final ID: TM2A.4

Demonstration of a high-finesse interferometer in attosecond science

Gabriele Ricci;[1];
1. The University of Texas at Austin, Austin, TX, United States.

Abstract (35 Word Limit): We report a topological Bragg grating that augments quantum sensing, reaching a value below 5² W/decade in our setup. Results indicate a clear pathway toward practical deployment. Our approach reduces system complexity while preserving fidelity.

Final ID: TM2B.1

Demonstration of a tunable laser source in remote sensing

Okada Emi;[1]; *Kato Naoki;*[1]; *Kimura Sho;*[1];
1. Tokyo metropokitan university, Hino, Tokyo, Japan.

Abstract (35 Word Limit): We present measurements of a resonant beam combiner operating in the context of optical networking. We discuss the trade-offs governing the design and their implications. Characterization confirms agreement with the proposed model.

Final ID: TM2B.2

Cryogenic interferometer for frequency metrology

William Hall;[1];

1. Optoelectronics Research Centre, University of Southampton, Southampton, Hampshire, United Kingdom.

Abstract (35 Word Limit): We present measurements of a broadband frequency comb operating in the context of quantum sensing, reaching a value below 10⁻¹² W/decade in our setup. Our approach reduces system complexity while preserving fidelity.

Final ID: TM2B.3

Toward spatially-multiplexed LiDAR ranging using a fiber amplifier

Florian Lehmann;[1]; *Tamar Biton;*[1];
1. Technion, Modiin, Israel.

Abstract (35 Word Limit): We report a ultrafast soliton that augments biophotonics. Characterization confirms agreement with the proposed model. Results indicate a clear pathway toward practical deployment.

Final ID: TM2B.4

Programmable, low-noise nanobeam enabling neuromorphic computing

Yoon Soyeon;[1]; *Khalid Farouk;*[1]; *Kim Jaewon;*[1,2]; *Park Jisoo;*[1]; *Jang Hana;*[3]; *Ahn Jihun;*[1]; *Lim Soyeon;*[4]; *Justin Garcia;*[1];

1. Department of Physics, Ulsan National Institute of Science and Technology, Ulsan, Korea (the Republic of).
2. Department of Mechanical Engineering, Pohang University of Science and Technology, Pohang, Korea (the Republic of).
3. Department of Physics, Chungbuk National University, Cheongju, Korea (the Republic of).
4. SLAC National Accelerator Laboratory, Menlo Park, CA, United States.

Abstract (35 Word Limit): This work introduces a topological ring resonator as a platform for coherent detection. Characterization confirms agreement with the proposed model.

Final ID: TM2C.1

Chip-scale, programmable optomechanical oscillator enabling LiDAR ranging

Zhao Na;[1];
1. Department of Physics, Korea University, Seoul, Korea (the Republic of).

Abstract (35 Word Limit): Here we investigate a chip-scale soliton suitable for quantum sensing. The device achieves competitive performance across the operating band.

Final ID: TM2C.2

Toward gain-switched quantum sensing using a photonic crystal

Dimitra Christodoulou;[1]; *Sara Ricci;*[1]; *Rachel Carter;*[1];
1. Physics, University of Toronto, Toronto, ON, Canada.

Abstract (35 Word Limit): We report a programmable thin-film platform that facilitates optical networking. The measured figures of merit exceed prior comparable demonstrations. Characterization confirms agreement with the proposed model.

Final ID: TM2C.3

Low-noise, tunable laser source enabling optical networking

Henry Wright;[1]; *Hao Kuo;*[1]; *Mary Carter;*[1]; *Freya Wood;*[1]; *Sara Gallo;*[1]; *Saoirse Kelly;*[1]; *Isabel Diaz;*[1];
Ruby Robinson;[1];
1. University of Cambridge, Cambridge, United Kingdom.

Abstract (35 Word Limit): We present measurements of a quantum-limited metasurface operating in the context of precision timing. Our approach reduces system complexity while preserving fidelity. The device achieves competitive performance across the operating band.

Final ID: TM2C.4

Toward few-cycle remote sensing using a metasurface

Ramesh Agarwal;[1]; *Hassan Al-Farsi;*[1]; *Sarah Martin;*[1]; *Aarav Desai;*[1];
1. University of Virginia, Charlottesville, VA, United States.

Abstract (35 Word Limit): Here we investigate a chip-scale soliton suitable for LiDAR ranging. The device achieves competitive performance across the operating band. These findings motivate further study of the underlying dynamics.

Final ID: TM2D.1

Demonstration of a dissipative photodetector in frequency metrology

Yoo Seojun;[1]; *Kim Hyeon;*[1];
1. Department of Physics, Pohang University of Science and Technology (POSTECH), Pohang, Gyeongsangbuk-do, Korea (the Republic of).

Abstract (35 Word Limit): This work introduces a resonant photonic crystal as a platform for quantum sensing, reaching a value below 2³ s/decade in our setup. Results indicate a clear pathway toward practical deployment.

Final ID: TM2D.2

Toward self-referenced frequency metrology using a photodetector

Matthew Thompson;[1]*; Mario Gil;*[1]*; William Ryan;*[1]*; Lily Hughes;*[1]*; Gilad Friedman;*[1]*; Ava Williams;*[1]*;*
Ava McCarthy;[1]*; Megan Wilson;*[1]*;*
1. Sandia National Laboratories, Albuquerque, NM, United States.

Abstract (35 Word Limit): We report a monolithic photonic crystal that enables quantum key distribution. Characterization confirms agreement with the proposed model.

Final ID: TM2D.3

Wavelength-agile, programmable interferometer enabling frequency metrology

Federico Lombardi; [1];
1. Information and Communications Technology, The University of Osaka, Suita, Osaka, Japan.

Abstract (35 Word Limit): This work introduces a coherent optical cavity as a platform for photonic computing. Our approach reduces system complexity while preserving fidelity.

Final ID: TM2D.4

Toward scalable precision timing using a interferometer

Ma Jing;[1,2]; *Hu Min;*[1,2]; *Chun Cheng;*[1,2]; *Huang Zhen;*[1,2]; *Gao Yu;*[1,2]; *Sun Hao;*[1,2]; *Sun Qiang;*[1,2];
1. State Key Laboratory of Information Photonics and Optical Communications, Beijing University of Posts and Telecommunications, Beijing, Beijing, China.
2. School of Electronic Engineering, Beijing University of Posts and Telecommunications, Beijing, Beijing, China.

Abstract (35 Word Limit): We present measurements of a chip-scale laser source operating in the context of quantum key distribution. The device achieves competitive performance across the operating band.

Final ID: SYMP1.1

Toward ultrafast attosecond science using a thin-film platform

Kai Hsu;[1];
1. University of Texas at Austin, Austin, TX, United States.

Abstract (35 Word Limit): We report a cryogenic thin-film platform that leverages frequency metrology. We discuss the trade-offs governing the design and their implications.

Final ID: SYMP1.2

Broadband gain medium for biophotonics

Goto Satoshi;[1]; *Watanabe Satoshi;*[1]; *Zhao Hao;*[2,3]; *Watanabe Yui;*[1]; *Hashimoto Yusuke;*[1]; *Kimura Yui;*[1]; *Yousef Mansour;*[1];

1. Graduate School of Natural Science and Technology, Shimane University, Matsue, Japan.
2. School of Engineering, Kogakuin University, Hachioji, Tokyo, Japan.
3. Research Center for Advanced Science and Technology, the University of Tokyo, Meguro, Tokyo, Japan.

Abstract (35 Word Limit): We present measurements of a wavelength-agile gain medium operating in the context of remote sensing, reaching a value below 2⁻¹² cm/decade in our setup. The device achieves competitive performance across the operating band. Results indicate a clear pathway toward practical deployment.

Final ID: SYMP1.3

Scalable quantum emitter for spectroscopy

Wen Tseng;[1]; *Shin Donghyun;*[2]; *Patricia Carter;*[3]; *Shin Jihun;*[2]; *Suresh Chatterjee;*[1];
1. Mechanical Science & Engineering, University of Illinois, Urbana, IL, United States.
2. Electrical & Computer Engineering, University of Illinois, Urbana, IL, United States.
3. Physics, University of Illinois, Urbana, IL, United States.

Abstract (35 Word Limit): We report a electrically-pumped gain medium that augments biophotonics. Characterization confirms agreement with the proposed model. We discuss the trade-offs governing the design and their implications.

Final ID: JTUP.1

Hybrid, dissipative quantum emitter enabling optical communication

Zhou Feng;[1];
1. Tsinghua University, Shenzhen, GUANGDONG, China.

Abstract (35 Word Limit): This work introduces a wavelength-agile photodetector as a platform for precision timing. These findings motivate further study of the underlying dynamics.

Final ID: JTUP.2

Toward hybrid photonic computing using a feedback loop

Declan Kelly; [1,2];
1. Joint Quantum Institute, UMD, College Park, MD, United States.
2. MND, NIST, Gaithersburg, MD, United States.

Abstract (35 Word Limit): We report a dispersion-engineered gain medium that supports quantum key distribution. Results indicate a clear pathway toward practical deployment.

Final ID: JTUP.3

Wavelength-agile, gain-switched entangled-photon pair enabling optical communication

Suzuki Satoshi;[1];
1. Tokushima University, Tokushima, TOKUSHIMA, Japan.

Abstract (35 Word Limit): A wavelength-agile Bragg grating is demonstrated for mid-infrared generation. The device achieves competitive performance across the operating band. Results indicate a clear pathway toward practical deployment.

Final ID: JTUP.4

Spatially-multiplexed, single-photon microresonator enabling remote sensing

Freddie McCarthy;[1,2]; *Liam Jones;*[1]; *Sanne Wouters;*[1]; *Ting Liu;*[1];
1. Physics, Harvard University, Cambridge, MA, United States.
2. Materials Science and Engineering, Boston University, Cambridge, MA, United States.

Abstract (35 Word Limit): Here we investigate a gain-switched beam combiner suitable for optical communication, reaching a value below 5⁻¹² s/decade in our setup. The device achieves competitive performance across the operating band. Characterization confirms agreement with the proposed model.

Final ID: JTUP.5

Low-noise Bragg grating for mid-infrared generation

Ananya Mukherjee;[1];
1. University of Rochester, Rochester, NY, United States.

Abstract (35 Word Limit): This work introduces a scalable photonic crystal as a platform for attosecond science. The measured figures of merit exceed prior comparable demonstrations.

Final ID: JTUP.6

Toward low-noise quantum key distribution using a nanobeam

Riccardo Galli;[1]; *Luca Rinaldi;*[1];
1. Research, Single Quantum, Delft, Netherlands.

Abstract (35 Word Limit): A resonant Bragg grating is demonstrated for optical communication. The measured figures of merit exceed prior comparable demonstrations.

Final ID: JTUP.7

Dispersion-engineered thin-film platform for mid-infrared generation

Antonio Leone;[1]; *Bram Claes;*[3]; *Faisal Rahman;*[1]; *William Smith;*[1]; *Luca Ferrari;*[2];
1. Univ. Grenoble Alpes, Univ. Savoie Mont Blanc, CNRS, Grenoble INP, CROMA, Grenoble, France.
2. Photonics Research Group, Dept. of Information Technology, Ghent University-imec, Ghent, Belgium.
3. Security in telecommunications, TU Berlin, Berlin, Germany.

Abstract (35 Word Limit): We report a spatially-multiplexed fiber amplifier that stabilizes photonic computing. The device achieves competitive performance across the operating band.

Final ID: JTUP.8

Toward self-referenced neuromorphic computing using a gain medium

Wu Bin;[1]; *Peng Kun;*[1];
1. Stanford, Mountain View, CA, United States.

Abstract (35 Word Limit): This work introduces a single-photon spectrometer as a platform for remote sensing, reaching a value below 5³ cm/decade in our setup. Our approach reduces system complexity while preserving fidelity. Results indicate a clear pathway toward practical deployment.

Final ID: JTUP.9

Quantum-limited, scalable fiber amplifier enabling optical networking

Oliver Robinson;[1,2]; *Oliver Patterson;*[1];
1. University of California at Berkeley, Berkeley, CA, United States.
2. JST PRESTO, Saitama, Japan.

Abstract (35 Word Limit): We report a integrated frequency comb that supports neuromorphic computing. These findings motivate further study of the underlying dynamics.

Final ID: JTUP.10

Coherent, spatially-multiplexed saturable absorber enabling beam steering

Freya Taylor;[1]; *Elias Werner;*[1];
1. Sandia National Laboratories, Albuquerque, NM, United States.

Abstract (35 Word Limit): This work introduces a tunable photodetector as a platform for attosecond science. Our approach reduces system complexity while preserving fidelity. These findings motivate further study of the underlying dynamics.

Final ID: JTUP.11

Toward chip-scale attosecond science using a supercontinuum

Ahmed Farouk;[1]; *Bram Claes;*[2];
1. Norfolk State University, Norfolk, VA, United States.
2. Lawrence Livermore National Laboratory, Livermore, CA, United States.

Abstract (35 Word Limit): We report a monolithic microresonator that enables quantum sensing. Our approach reduces system complexity while preserving fidelity.

Final ID: JTUP.12

Gain-switched, high-finesse waveguide lattice enabling mid-infrared generation

Varun Patel;[1]; *Neha Naidu;*[2]; *Sanjay Gupta;*[1];
1. TCS Research, Bangalore, Karnataka, India.
2. Indian Institute of Technology Bombay, Mumbai, India.

Abstract (35 Word Limit): This work introduces a single-photon entangled-photon pair as a platform for neuromorphic computing. We discuss the trade-offs governing the design and their implications. Our approach reduces system complexity while preserving fidelity.

Final ID: WM1A.1

Demonstration of a electrically-pumped metasurface in coherent detection

Yousef Habib;[1]; *Freddie Thompson;*[1]; *Agnieszka Novak;*[2,3]; *Anita Kaminski;*[2,3]; *Saoirse Evans;*[1];
Istvan Szymanski;[1]; *Federico Greco;*[1,4];
1. Photonics Institute, TU Wien, Vienna, Austria.
2. Lukasiewicz Institute of Microelectronics and Photonics, Warsaw, Poland.
3. Faculty of Physics, University of Warsaw, Warsaw, Poland.
4. Institute of Chemistry and Environmental Sciences,University of Ss. Cyril and Metodius, Trnava, Slovakia.

Abstract (35 Word Limit): We report a dissipative gain medium that supports quantum key distribution. Results indicate a clear pathway toward practical deployment.

Final ID: WM1A.2

Demonstration of a coherent photonic crystal in biophotonics

<u>*Faisal Habib;*</u>[1]; *Robert Martinez;*[2]; *Lucia Ruiz;*[3]; *Ma Lei;*[1]; *Liang Ping;*[4]; *Hu Feng;*[1]; *Sanne van Dijk;*[1]; *Faisal Saleh;*[5];

1. Electrical Engineering, California Institute of Technology, Pasadena, CA, United States.
2. Department of Electrical Engineering and Computer Science, Howard University, Washington, D.C., DC, United States.
3. Applied Physics and Material Science, California Institute of Technology, Pasadena, CA, United States.
4. Physics, Mathematics, and Astronomy, California Institute of Technology, Pasadena, CA, United States.
5. Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA, United States.

Abstract (35 Word Limit): A spatially-multiplexed soliton is demonstrated for spectroscopy, reaching a value below 10⁻⁹ cm/decade in our setup. Our approach reduces system complexity while preserving fidelity. Results indicate a clear pathway toward practical deployment.

Final ID: WM1A.3

Toward wavelength-agile biophotonics using a interferometer

Grace Jones;[1]; *Ruby Wood;*[1,2]; *Thomas Patterson;*[1,2]; *Rocio Diaz;*[3]; *Pieter Vermeulen;*[4]; *Kevin Phillips;*[5]; *Maryam Hassan;*[1,2];

1. Department of Electrical, Computer, and Energy Engineering, University of Colorado Boulder, Boulder, CO, United States.
2. Department of Physics, University of Colorado Boulder, Boulder, CO, United States.
3. Department of Cell and Developmental Biology, University of Colorado Anschutz Medical Campus, Aurora, CO, United States.
4. Department of Physics and Astronomy, University of Denver, Denver, CO, United States.
5. Department of Bioengineering, University of Colorado Anschutz Medical Campus, Aurora, CO, United States.

Abstract (35 Word Limit): Here we investigate a ultrafast photodetector suitable for quantum sensing. We discuss the trade-offs governing the design and their implications.

Final ID: WM1A.4

Demonstration of a thermally-tuned nanobeam in mid-infrared generation

Tang Yu;[1]; *Han Hui;*[1]; *Cheng Lai;*[2]; *Okada Yui;*[2]; *Nakamura Haruki;*[2]; *Shu Chang;*[1];
1. Graduate Institute of Photonics and Optoelectronics, and Department of Electrical Engineering, National Taiwan University, Taipei, Taiwan.
2. National Institute of Information and Communications Technology, Tokyo, Japan.

Abstract (35 Word Limit): A chip-scale entangled-photon pair is demonstrated for remote sensing. Results indicate a clear pathway toward practical deployment. Characterization confirms agreement with the proposed model.

Final ID: WM1B.1

Thermally-tuned optical cavity for quantum key distribution

Hong Chang;[1]; *Riccardo Mancini;*[1,2]; *Pei Kuo;*[1]; *Ma Fang;*[1,2]; *Xie Peng;*[3]; *Jia Lee;*[1]; *Miguel Reyes;*[1,3]; *Wen Yang;*[1];

1. Institute of Microelectronics, A*STAR (Agency for Science, Technology and Research), Singapore, Singapore.
2. Center for Intelligent Sensors and MEMS, Department of Electrical and Computer Engineering, National University of Singapore, Singapore, Singapore.
3. National Semiconductor Translation and Innovation Centre (NSTIC), Singapore, Singapore.

Abstract (35 Word Limit): Here we investigate a programmable beam combiner suitable for nonlinear microscopy, reaching a value below 2⁻¹² W/decade in our setup. Our approach reduces system complexity while preserving fidelity.

Final ID: WM1B.2

Resonant, monolithic ring resonator enabling attosecond science

<u>*Lily Thompson;*</u>[1,3]; *Allison Jones;*[1]; *Charlotte O'Brien;*[1]; *Bilal Said;*[2]; *Linda Carter;*[1];

1. Wyant College of Optical Sciences, Tucson, AZ, United States.
2. Pacific Northwest National Labs, Richland, WA, United States.
3. Physics, University of Arizona, Tucson, AZ, United States.

Abstract (35 Word Limit): We present measurements of a ultrafast thin-film platform operating in the context of terahertz imaging. The measured figures of merit exceed prior comparable demonstrations.

Final ID: WM1B.3

Self-referenced interferometer for coherent detection

<u>Yousef Said;</u>[1]; *Suzuki Sota;*[1]; *Thomas Davies;*[1]; *Huang Tao;*[1]; *Eitan Adler;*[1]; *Feng Yan;*[1]; *Lukas Kovacs;*[2]; *Mary Anderson;*[1];

1. Applied and Engineering Physics, Cornell University, Pittsburgh, PA, United States.
2. E. L. Ginzton Laboratory, Stanford University, Stanford, CA, United States.

Abstract (35 Word Limit): We report a electrically-pumped metasurface that stabilizes photonic computing, reaching a value below 2⁻¹² cm/decade in our setup. The measured figures of merit exceed prior comparable demonstrations.

Final ID: WM1C.1

Scalable, reconfigurable quantum emitter enabling coherent detection

Allison Baker;[1];
1. Institue for Energy Efficiency, UCSB, Santa Barbara, CA, United States.

Abstract (35 Word Limit): A spatially-multiplexed interferometer is demonstrated for frequency metrology. These findings motivate further study of the underlying dynamics.

Final ID: WM1C.2

Demonstration of a quantum-limited ring resonator in optical communication

Declan Green;[1]; *Niall Murphy;*[1]; *Ma Long;*[1]; *Yu Wu;*[1]; *Daniel Phillips;*[1]; *Nikolaos Spanos;*[1]; *Eric Clark;*[1]; *Bilal Aziz;*[1];

1. ECE, University of Virginia, Charlottesville, VA, United States.

Abstract (35 Word Limit): We report a single-photon interferometer that supports nonlinear microscopy, reaching a value below 5⁻¹² cm/decade in our setup. The device achieves competitive performance across the operating band.

Final ID: WM1C.3

Toward ultrafast neuromorphic computing using a phased array

Sigrid Lund;[1];
1. Physics, Harvard University, Cambridge, MA, United States.

Abstract (35 Word Limit): Here we investigate a gain-switched interferometer suitable for quantum key distribution. These findings motivate further study of the underlying dynamics. Characterization confirms agreement with the proposed model.

Final ID: WM1C.4

Topological nanobeam for mid-infrared generation

Fujita Hiroshi;[1]; *Nakamura Rina;*[1];
1. Physics, Keio University, Kohoku-ku, Yokohama, Kanagawa, Japan.

Abstract (35 Word Limit): Here we investigate a self-referenced optical cavity suitable for coherent detection. Results indicate a clear pathway toward practical deployment. The device achieves competitive performance across the operating band.

Final ID: WM1D.1

Dispersion-engineered, phase-stable optical cavity enabling biophotonics

Omar Younis;[1]; *Divya Singh;*[1];
1. Indian Institute of Science (IISc).

Abstract (35 Word Limit): A single-photon interferometer is demonstrated for quantum sensing, reaching a value below 5² cm/decade in our setup. Our approach reduces system complexity while preserving fidelity.

Final ID: WM1D.2

Coherent, resonant spectrometer enabling mid-infrared generation

Agnieszka Molnar;[1]; *Isla Brown;*[1]; *Ville Sandberg;*[1]; *Elias Frank;*[1]; *William Thompson;*[1]; *Wouter Aerts;*[1]; *Daan van Dijk;*[1];
1. University of Colorado Boulder, Boulder, CO, United States.

Abstract (35 Word Limit): We present measurements of a coherent frequency comb operating in the context of terahertz imaging. Our approach reduces system complexity while preserving fidelity.

Final ID: WM1D.3

Demonstration of a programmable gain medium in spectroscopy

Peng Feng;[1]; *Deng Yan;*[1];
1. Department of Electronic Engineering, The Chinese University of Hong Kong, Shatin, New Territories, Hong Kong.

Abstract (35 Word Limit): We present measurements of a gain-switched quantum emitter operating in the context of biophotonics, reaching a value below 10⁻⁹ cm/decade in our setup. These findings motivate further study of the underlying dynamics.

Final ID: WM1D.4

Demonstration of a resonant ring resonator in biophotonics

Francesco Galli;[1];
1. Sapienza Universita di Roma.

Abstract (35 Word Limit): We present measurements of a dissipative thin-film platform operating in the context of coherent detection. We discuss the trade-offs governing the design and their implications. These findings motivate further study of the underlying dynamics.

Final ID: WM2A.1

Electrically-pumped beam combiner for coherent detection

Sasaki Ryo;[1,2]; *Yamaguchi Aoi;*[2]; *Jia Yang;*[2]; *Okada Yuki;*[1]; *Kimura Keisuke;*[1];
1. Department of Energy Engineering, Nagoya University, Nagoya, Japan.
2. Photonics-Electronics Integration Research Center, National Institute of Advanced Industrial Science and Technology, Tsukuba, Japan.

Abstract (35 Word Limit): Here we investigate a ultrafast phased array suitable for optical communication. We discuss the trade-offs governing the design and their implications.

Final ID: WM2A.2

Demonstration of a scalable waveguide lattice in LiDAR ranging

Michal Dvorak;[1]; *Feng Zhen;*[1]; *Carlos Fernandez;*[1]; *Giuseppe Lombardi;*[2]; *Lorenzo Romano;*[1]; *Ewan Williams;*[1]; *Florian Dubois;*[1]; *Dimitra Vlachos;*[1];

1. University of Southampton, Southampton, United Kingdom.
2. University of Pavia, Pavia, Italy.

Abstract (35 Word Limit): This work introduces a single-photon beam combiner as a platform for mid-infrared generation. We discuss the trade-offs governing the design and their implications.

Final ID: WM2A.3

Demonstration of a nonlinear interferometer in precision timing

Pei Lee;[1,2]; *Hao Yang;*[1,2]; *Wang Yang;*[1,2]; *Zhu Bin;*[1,2]; *Hsin Wu;*[1,2]; *Wen Yang;*[1,2]; *Deng Lei;*[1,2];
1. Institute of Advanced Photonics Technology, School of Information Engineering, Guangdong University of Technology, Guangzhou, China.
2. Key Laboratory of Photonic Technology for Integrated Sensing and Communication, Ministry of Education of China, Guangdong University of Technology, Guangzhou, China.

Abstract (35 Word Limit): Here we investigate a electrically-pumped ring resonator suitable for remote sensing. The measured figures of merit exceed prior comparable demonstrations. Characterization confirms agreement with the proposed model.

Final ID: WM2B.1

Spatially-multiplexed, thermally-tuned optomechanical oscillator enabling beam steering

Arjun Menon;[1,2]; *Rohan Kumar;*[1]; *Aarav Rao;*[1];
1. Metallurgical and Materials Engineering, Indian Institute of Technology Ropar, Rupnagar, Punjab, India.
2. Metallurgical and Materials Engineering, Punjab Engineering College (Deemed to be University), Sector12, Chandigarh, UT Chandigarh, India.

Abstract (35 Word Limit): Here we investigate a high-finesse laser source suitable for photonic computing. We discuss the trade-offs governing the design and their implications. The measured figures of merit exceed prior comparable demonstrations.

Final ID: WM2B.2

Reconfigurable entangled-photon pair for attosecond science

Sem de Boer;[1]; *Jack Thomas;*[1]; *Sophie Green;*[1]; *Mary Thomas;*[1]; *Liam Robinson;*[1]; *Sophie Thomas;*[1];
Davide Barbieri;[1]; *Niall Williams;*[1];
1. University of Southampton, Southampton, United Kingdom.

Abstract (35 Word Limit): We report a scalable nanobeam that supports LiDAR ranging. Results indicate a clear pathway toward practical deployment. We discuss the trade-offs governing the design and their implications.

Final ID: WM2B.3

Demonstration of a nonlinear optical cavity in spectroscopy

Xie Ping;[1]; *Feng Bo;*[1]; *He Lei;*[2]; *Jia Chiang;*[1]; *Feng Yang;*[1]; *Ting Liu;*[2]; *Zhang Long;*[2]; *Sun Fang;*[1];
1. Department of Electronic Engineering, Shanghai Jiao Tong University, Shanghai, China.
2. School of Physics and Astronomy, Shanghai Jiao Tong University, Shanghai, China.

Abstract (35 Word Limit): Here we investigate a reconfigurable microresonator suitable for quantum key distribution. The measured figures of merit exceed prior comparable demonstrations. Characterization confirms agreement with the proposed model.

Final ID: WM2C.1

Demonstration of a thermally-tuned photonic crystal in quantum sensing

Kwon Jiwoo;[1].
1. QIST, Seoul, Korea (the Republic of).

Abstract (35 Word Limit): We report a monolithic optomechanical oscillator that underpins LiDAR ranging. Characterization confirms agreement with the proposed model. Characterization confirms agreement with the proposed model.

Final ID: WM2C.2

Self-referenced, resonant beam combiner enabling frequency metrology

Wang Jing;[1]; *Gao Hui;*[1]; *Alice Moretti;*[1]; *Liam Wright;*[1]; *Yu Lai;*[2]; *Ahmed Khalil;*[1]; *Yousef Nasser;*[1];
1. Electrical and Computer Engineering, University of Florida, Gainesville, FL, United States.
2. Department of Mechanical Engineering, National Taiwan University, Taipei, Taiwan.

Abstract (35 Word Limit): We present measurements of a gain-switched optical cavity operating in the context of coherent detection. These findings motivate further study of the underlying dynamics. The device achieves competitive performance across the operating band.

Final ID: WM2C.3

Toward topological nonlinear microscopy using a phased array

Amelia Edwards;[1]; *Li Ping;*[1]; *Zhou Long;*[1,2]; *Olga Kozlov;*[3,4]; *Joshua Anderson;*[3]; *Richard Taylor;*[5]; *Huang Shan;*[5]; *Jiri Marek;*[1];

1. University of Michigan, Ypsilanti, MI, United States.
2. Lawrence Berkeley National Laboratory, Berkeley, CA, United States.
3. Cornell University, Ithaca, NY, United States.
4. Technion-IIT, Haifa, Israel.
5. Clemson University, Anderson, SC, United States.

Abstract (35 Word Limit): We report a spatially-multiplexed optomechanical oscillator that facilitates remote sensing. Our approach reduces system complexity while preserving fidelity. Results indicate a clear pathway toward practical deployment.

Final ID: WM2D.1

Demonstration of a broadband nanobeam in quantum key distribution

Patricia Young;[1]; *Thomas Wood;*[1]; *Rocio Rodriguez;*[1]; *Amira Younis;*[1]; *Meera Choudhury;*[1]; *Ivan Pavlov;*[1];
1. Massachusetts Institute of Technology.
2. Tecnologico de Monterrey.

Abstract (35 Word Limit): We present measurements of a gain-switched gain medium operating in the context of coherent detection. Our approach reduces system complexity while preserving fidelity.

Final ID: WM2D.2

Demonstration of a resonant feedback loop in attosecond science

Tariq Ibrahim;[1]; *Cai Kun;*[1]; *Cai Long;*[2]; *Chih Wu;*[1]; *Wang Jing;*[2]; *Han Jing;*[3];
1. Zhejiang Lab, Hangzhou, China.
2. Guangxi Medical University, Nanning, China.
3. Shenzhen University of Information Technology, Shenzhen, China.

Abstract (35 Word Limit): We present measurements of a resonant thin-film platform operating in the context of nonlinear microscopy. Our approach reduces system complexity while preserving fidelity.

Final ID: WM2D.3

Toward high-finesse biophotonics using a frequency comb

Elena Alonso;[1]; *Sophie White;*[1]; *Lieke Peeters;*[1];
1. Eindhoven University of Technology, Eindhoven, Netherlands.

Abstract (35 Word Limit): We present measurements of a dispersion-engineered saturable absorber operating in the context of optical networking. The device achieves competitive performance across the operating band. These findings motivate further study of the underlying dynamics.

Final ID: WT1.1

Nonlinear supercontinuum for photonic computing

Nathan Wright;[1]; *Aoife Robinson;*[1]; *Lukas Bauer;*[1];
1. Lawrence Livermore National Laboratory, Livermore, CA, United States.

Abstract (35 Word Limit): We report a polarization-resolved supercontinuum that accelerates biophotonics, reaching a value below 2³ cm/decade in our setup. These findings motivate further study of the underlying dynamics. The device achieves competitive performance across the operating band.

Final ID: WT1.2

Demonstration of a mode-locked spectrometer in photonic computing

Freddie MacLeod;[1];
1. Yale University, New Haven, CT, United States.

Abstract (35 Word Limit): We present measurements of a single-photon waveguide lattice operating in the context of precision timing. The measured figures of merit exceed prior comparable demonstrations.

Final ID: WT1.3

Toward chip-scale optical communication using a optomechanical oscillator

Noura Rahman;[1,2]; *Lily Evans;*[3]; *Matthew Jones;*[3]; *Olivia Brown;*[1,2]; *Kabir Bhat;*[1,2];

1. JQI, NIST/UMD, College Park, MD, United States.
2. MND, NIST, Gaithersburg, MD, United States.
3. AIM Photonics, Albany, NY, United States.

Abstract (35 Word Limit): Here we investigate a low-noise phased array suitable for beam steering. We discuss the trade-offs governing the design and their implications. Our approach reduces system complexity while preserving fidelity.

Final ID: PD1.1

Toward coherent photonic computing using a Bragg grating

Guo Shan;[1]; *Pei Liu;*[1]; *Luo Tao;*[1]; *Yu Lai;*[1]; *Xu Hui;*[1]; *Gao Yan;*[1]; *Zhang Wei;*[1];
1. Huazhong University of Science and Technology, Wuhan, China.

Abstract (35 Word Limit): Here we investigate a resonant nanobeam suitable for terahertz imaging, reaching a value below 5² W/decade in our setup. The measured figures of merit exceed prior comparable demonstrations. These findings motivate further study of the underlying dynamics.

Final ID: PD1.2

Few-cycle, broadband spectrometer enabling LiDAR ranging

Callum Davies;[1]; *Jonas Braun;*[1]; *Grace Thompson;*[1];
1. Electrical and Computer Engineering, Georgia Institute of Technology, Atlanta, GA, United States.

Abstract (35 Word Limit): This work introduces a nonlinear quantum emitter as a platform for biophotonics, reaching a value below 10⁻¹² W/decade in our setup. Results indicate a clear pathway toward practical deployment.

Final ID: PD2.1

Gain-switched, nonlinear ring resonator enabling beam steering

Noura Nasser;[1]; *Zhu Yan;*[1]; *Liang Kun;*[1]; *Simone Conti;*[1]; *Fatima Saleh;*[1];
1. The City University of New York, New York, NY, United States.

Abstract (35 Word Limit): This work introduces a dissipative fiber amplifier as a platform for photonic computing. Characterization confirms agreement with the proposed model. We discuss the trade-offs governing the design and their implications.

Final ID: PD2.2

Low-noise phased array for frequency metrology

Megan Rodriguez;[1]; *Lorenzo Mancini;*[1]; *Stephen Baker;*[1];
1. University of Toronto, Toronto, ON, Canada.

Abstract (35 Word Limit): A broadband photonic crystal is demonstrated for quantum key distribution. Characterization confirms agreement with the proposed model.